Clark Zhang

Email | LinkedIn | Portfolio

Work Authorization: Canadian Citizen | Eligible for U.S. TN Status (No H-1B required)

EDUCATION

University of British Columbia **September 2021 - April 2026**
Bachelor of Applied Science - Integrated Engineering (Mechatronics Specialization), Co-op Program
cGPA: 4.0 (92%)

- Relevant Courses: Mechanics III: 95%, Automatic Control Systems: 88%, Electromechanics: 94%, Thermodynamics: 100%, Heat Transfer, 94%, Fluid Mechanics: 91%

SKILLS

- Mechanical Design: SolidWorks (CSWP Certified + Simulations), CATIA/3DX, Fusion 360, AutoCAD
- Programming & Electronics: Python, C++, PLC Control (UR/Kuka), PCB design (KiCAD), ROS2, Git, MATLAB/Simulink
- Prototyping & Manufacturing: FDM/SLA 3D printing, CNC machining, laser/waterjet cutting

WORK EXPERIENCE

Hein Labs **December 2024 – January 2026**
Engineering Project Lead - Mechatronics | SolidWorks, Fusion 360, GD&T, ROS2, C++, Python, KiCAD

- Directed a **6-person engineering team** developing a robotic drug-analysis platform for BC's overdose crisis response, automating gravimetric analysis, liquid transfer, centrifugation, and downstream analysis with **98% overall reliability**.
- Engineered and manufactured **3D-printable lab automation modules** in **Fusion 360/SolidWorks**, including a BLDC centrifuge, syringe-pump liquid handler, decapper, robotic vial gripper, and transfer fixtures, reducing module costs from **$1,500–$5,000 to <$100**.
- Iteratively prototyped custom hardware using **GD&T principles** to meet chemical-analysis requirements for precise fluid transfer, repeatable vial positioning, reliable decapping, and robotic sample handling.
- Developed a multi-tiered control architecture using **Python** to orchestrate **UR robotic arm motion** and high-level workflows, integrated with **C++/ROS2** for hardware module automation and computer-vision error detection.
- Designed and fabricated **2 custom PCBs** using **KiCAD** for synchronized stepper and BLDC motor control

Hein Labs **April 2024 – December 2024**
Mechatronics Engineering Intern | SolidWorks, RoboDK, Python, C++, Git, PLC, I/O Wiring

- Commissioned **4 industrial robotic arms** across **3 deployed self-driving chemistry lab platforms**, handling robot mounting, I/O wiring, networking, and end-to-end system integration while reducing scientists' hands-on time by **~80%**.
- Transitioned hard-to-maintain **PLC-based robot sequences** to a reusable **Python control library** for **UR and Kuka** arms, making workflows easier to program, modify, and deploy.
- Modeled **2 robotic platforms** in **SolidWorks** and simulated workflows in **RoboDK** to verify reachability, collision clearance, and lab module access before deployment, reducing workflow errors by **5x**.

UBC AeroDesign **April 2022 - April 2026**
Airfoils Team Lead | CATIA, SolidWorks, ANSYS, AutoCAD, GD&T, Manufacturing, Composites

- Led an **11-person subteam** designing and manufacturing a **120-inch heavy-lift RC aircraft** for SAE AeroDesign competition, overseeing wing design, control-surface integration, manufacturing, and testing. Earned 2nd place overall in 2025
- Designed main wing, ailerons, servo mounts, and linkage interfaces in **SolidWorks and CATIA V5** using **DFMA and GD&T principles**, improving alignment and manufacturability
- Optimized aircraft geometry through **SolidWorks Flow, ANSYS, FEA, and wind-tunnel validation**, improving lift performance by **~20%** and verifying structural stiffness before manufacturing
- Manufactured flight hardware using **3D printing, CNC machining, waterjet/laser cutting, and carbon-fiber prepreg layups**, standardizing assembly processes and reducing build errors by **70%**.

PROJECTS

Robotic Tool Vending Machine (Haven) **September 2025 - April 2026**
Project Lead | SolidWorks, Raspberry Pi, Klipper, KiCAD, Manufacturing & Prototyping

- Led design of a vertical **Cartesian XY gantry** and rotary storage system in **SolidWorks** for automated tool dispensing
- Fabricated and integrated prototypes using **3D printing, waterjet cutting, hand tools, and CNC machining**
- Developed **Raspberry Pi/Klipper motion control** with magnetic encoder feedback and **custom I2C multiplexer PCB** to synchronize gantry travel and storage indexing.

Semi-Autonomous Aquatic Trash-Removal Robot **September 2024 - April 2025**
Mechanical Lead | SolidWorks, 3D-printing, Waterjet, Machine Shop Tools

- Designed **aquatic mechanical assemblies** in **SolidWorks**, engineering custom dual-pontoon flotation chassis, structural supports, and adjustable propulsion mounts
- Fabricated and integrated **sealed plywood, aluminum, and 3D-printed components** to assemble a watertight product optimized for saltwater operations